

Appeals-NLP: Hierarchical Classification of Russian Spoken-Language Housing Maintenance Requests under ASR Noise

NLP & LLMs Course — Final Project

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Abstract

We study automatic classification of Russian-language housing-utility maintenance requests transcribed by a production voice robot (ASR). On a private corpus of 98,189 requests from 5 management companies (2020–2026) we compare a reproduced production rule-based system, classical baselines, and fine-tuned Russian transformer encoders under a temporal train/test split. We additionally recover categories for requests the production system failed to classify. Code: https://github.com/cashman2100/appeals_nlp.

1 Introduction

Dispatch services for residential buildings receive large volumes of requests by phone. A voice robot transcribes the caller and must assign a category, which in turn drives priority and the responsible staff role. Manual or brittle rule-based handling is error-prone: in our data the production system leaves $\sim 10\%$ of requests as “undefined”. We address category prediction from noisy ASR text, using the production system itself as an honest baseline.

Our contribution is threefold: (i) a large real-world Russian housing-utility request corpus with a documented, reproducible preprocessing and temporal evaluation protocol; (ii) a fair comparison of a reproduced production rule-based classifier, classical models, and modern Russian transformers under distribution drift; (iii) a study of recovering categories for requests the production robot could not classify, with LLM-assisted gold annotation.

1.1 Team

Solo project. Andrey Ivanov — data engineering, modelling, evaluation, and report.

2 Related Work

Russian pretrained encoders. Zmitrovich et al. [11] release a family of 13 Russian Transformer LMs (ruBERT, ruRoBERTa, ruT5), building on BERT [2] and RoBERTa [6]; ruRoBERTa-large attains the strongest RussianSuperGLUE [8] scores in its size class, motivating its use as our main encoder. Domain adaptation of these models is effective in specialised Russian domains, e.g. Ru-BioRoBERTa for biomedical text [10], and they remain strong embedding backbones [9].

Customer-support intent / ticket classification. Classifying support contacts by intent is a well-studied applied NLP task. Li et al. [4] move from multiclass to semi-supervised multi-task

learning over support text and report large gains, highlighting the value of multi-task formulations and of leveraging negative/unlabelled cases — directly relevant to our priority/performer/undefined structure. Moreno et al. [7] build a universal intent recogniser for service chatbots with multi-task learning over intents and entities. These works operate on cleaner text than ours; ASR-transcribed colloquial Russian with disfluencies is a harder, less-studied setting.

Class imbalance and evaluation. Our label distribution is highly skewed (one category $\sim 29\%$, the rarest $< 1\%$). Cost-sensitive training and focal loss [5] are standard remedies; adaptable focal-loss variants target imbalanced text specifically [1]. Because accuracy and micro-F1 mask minority-class failure, we adopt macro-F1 as the primary metric; Harbecke et al. [3] analyse exactly this choice of weighting scheme for imbalanced classification.

Positioning. Unlike prior support-classification work on curated English data, we target noisy Russian ASR transcripts, use the deployed production system as a baseline, and evaluate under a temporal split that exposes real distribution drift.

3 Model Description

3.1 Architecture

Our model follows a shared-encoder, pluggable-head design. The backbone is **ruRoBERTa-large** [11] (355M parameters, max position 512 tokens, vocabulary size 50 257), with **ruBERT-base** (179M) used for rapid pilot iteration. We chose the large variant because our v1 experiments (§3.3) showed **ruBERT-base** at statistical parity with TF-IDF classical baselines, while the larger model’s richer contextualisation of long colloquial ASR inputs proved decisive for minority-class performance.

Classification is performed by a linear projection $1024 \rightarrow K$ ($K=15$ full or $K=13$ consolidated), preceded by dropout ($p=0.2$) on the [CLS] token representation.

Pluggable heads. The architecture implements a **SharedEncoder + ModuleDict(heads)** pattern: the transformer encoder is shared, and multiple task-specific heads can be attached without re-training the backbone. In the current work only the **category** head is active (enabled via `tasks.category.enabled: true` in the YAML configuration). This design directly enables future multi-task learning of priority and performer heads (§7) without architectural changes, and was established as a hard constraint in the project architecture decisions.

3.2 Loss Function and Class Imbalance

The training objective combines focal loss [5] with per-class weights and label smoothing:

$$\mathcal{L} = (1 - p_t)^\gamma \text{CE}(y, \hat{y}; w_c, \alpha), \quad (1)$$

where p_t is the model probability for the true class, $\gamma = 2$ the focusing parameter, $w_c = N / (K n_c)$ the per-class weight, and $\alpha = 0.05$ the label smoothing coefficient.

Why $\gamma = 2$. This is the standard value from Lin et al. [5]; it reduces gradient weight for easily-classified examples (Heating, Electricity, Intercom, all > 0.93 F1) and amplifies learning on confused water-related cases and rare classes (Repair, Drainage clogs, both < 0.25 F1 in v1). Class weights w_c address the 60:1 long-tail imbalance (Heating vs. Repair). Label smoothing $\alpha=0.05$ mitigates overconfidence on noisy post-edit labels (see §7).

3.3 Hyperparameter Selection: v1 to v2

We iterated across two configurations, selecting on validation macro-F1.

v1 (initial): `max_length=128`, weighted cross-entropy (no focal), $\text{lr} = 2 \times 10^{-5}$, 3 epochs. Result: `ruRoBERTa-large` 0.646 macro-F1 — statistical parity with TF-IDF + LinearSVC (0.647, CI intervals overlap). Two root causes were identified: (i) `max_length=128` silently truncated 4.6% of requests longer than 150 characters — exactly the segment where contextual understanding should outperform bag-of-words features (§6.4); (ii) weighted cross-entropy alone was insufficient for the long tail (Repair, Drainage clogs $\text{F1} < 0.25$).

v2 (final): `max_length=256` (covers $>99\%$ of texts uncropped), focal loss $\gamma=2$ with label smoothing $\alpha=0.05$, $\text{lr} = 1.5 \times 10^{-5}$, up to 5 epochs with early stopping on validation macro-F1, fp16, on Kaggle T4. Result: 0.655 macro-F1 (full taxonomy) / 0.699 (consolidated), with confidence intervals disjoint from both TF-IDF and v1. The segment analysis (§6.4) confirms that the gain is concentrated precisely in the long-text segment that v1 truncated.

3.4 Production Rule-Based Baseline

The reproduced baseline re-implements the voice robot’s classification logic: a longest-match lookup over the production synonym table (≈ 600 synonym phrases across 24 categories). For each request, all synonyms are matched against the lowercased text; the match spanning the most characters determines the category, with ties broken by category frequency; requests with no match are labelled “undefined”. The baseline is reproducible via `src/models/robot_baseline.py`.

Caveat. The synonym table reflects today’s production configuration. Historical requests (2020–2022) may have been classified by older rule versions, making this a lower bound on the system’s historical accuracy (§7).

4 Dataset

The corpus contains 98,189 robot-created requests from 5 management companies spanning 2020–2026, in Russian, transcribed by the production ASR voice robot. Transcription characteristics: 74% fully lowercase text, 16% containing punctuation marks, and a median request length of 42 characters. 19% of requests are exact duplicates, reflecting repeated callers reporting the same persistent issue. Labels are post-edit production categories — the original robot choices are not logged (see §7).

Working taxonomy: top-14 child categories + “Other” (rare $<1\%$ classes folded); “undefined” (10%) is held out as a reclassification pool (§6.3). Split is *temporal* (train 2020–2024 / test 2025–2026), imitating deployment. Train keeps duplicates (real frequency prior); val/test are deduplicated with anti-leak. Full statistics and procedure: `data/DATACARD.md`. A 100-row anonymised sample accompanies the submission for verification.

4.1 Category Distribution

Figure 1 shows the label distribution across the 15-class working taxonomy. The distribution is strongly skewed: Heating dominates at $\sim 29\%$ of labelled requests, while Repair, the rarest working category, accounts for under 0.5% — a 60:1 ratio. This severe imbalance makes per-class macro-F1 both critical and challenging; accuracy and micro-F1 would mask systematic failure on minority classes [3].

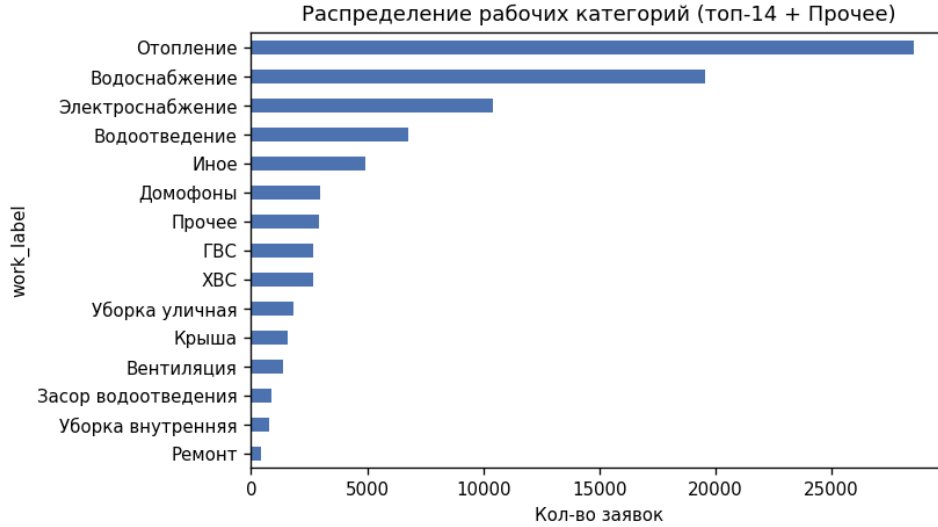


Figure 1: Category distribution (15-class taxonomy, labelled requests only). Heating dominates; Repair and Drainage clogs are the rarest working categories (60:1 imbalance ratio).

4.2 ASR Noise Characteristics

Figure 2 shows the request length distribution. Texts are short (median 42 chars) but span a wide range: 22% are very short (<25 characters, amenable to keyword lookup) while 4.6% are long (>150 characters), comprising “rant”-style narrations that require contextual understanding. Approximately 0.2% of requests contain profane language reflecting high caller frustration.

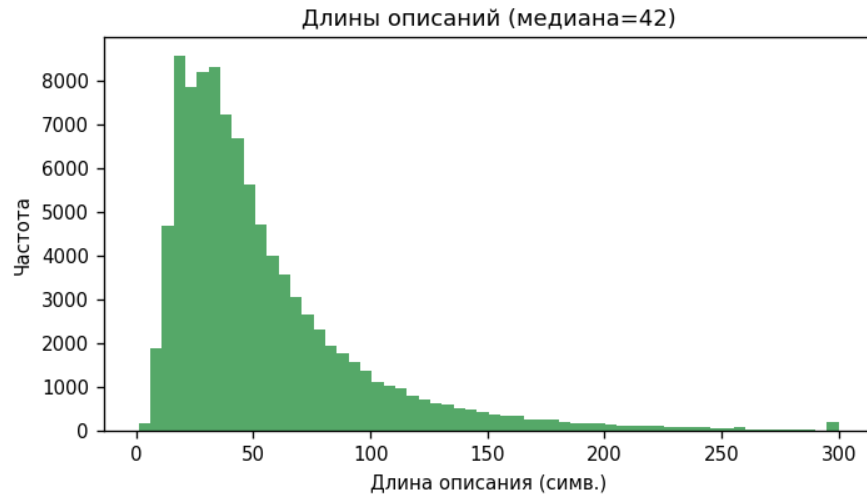


Figure 2: Distribution of request lengths (characters; linear scale, clipped at 300). Short texts dominate; the long tail of “rant”-style requests is the hardest segment for classification.

Table 1 illustrates the diversity of ASR-transcribed colloquial Russian in this corpus. The examples cover phenomena that make classification harder than standard NLP benchmarks: absent punctuation, spoken word order, ASR disfluencies (word repetitions, broken boundaries), telegraphic

omissions, emotional register, and cross-category lexical ambiguity.

Table 1: Representative ASR request texts with annotated linguistic phenomena. Category names are translated; original Russian shown verbatim.

Request text	Phenomenon	True category
нет тепла трёхкомнатная квартира только в одной комнате тепло	Long colloquial phrase, no punctuation	Heating
холодная батарея	Telegraphic	Heating
нет воды	Lexically am- biguous (hot? cold? general?)	Water supply
засор у нас засор везде вода из квартиры	ASR repeti- tions, rant style	Drainage
после включения отопления опять течёт с потолка	Contextual ref- erence misleads key- word model	Heating
тепло у меня нет	Non-standard word order	UNDEFINED

4.3 Temporal Split and Distribution Drift

Figure 3 shows the monthly volume of Heating requests, illustrating strong seasonal structure. The temporal split (train ≤ 2024 / test ≥ 2025) therefore exposes real distribution drift rather than a random data partition. Table 2 quantifies class-share changes between periods. The largest shifts are in Street cleaning (+1.9 pp, +119% relative), Roof (+2.0 pp, +167% relative), and Drainage (+2.3 pp, +32% relative), reflecting seasonal maintenance patterns and changes in service scope across the companies. Random splitting would mask this drift and produce optimistically biased generalisation estimates.

Table 2: Class-share drift: train (≤ 2024) vs. test (≥ 2025), consolidated taxonomy. Top 6 categories by absolute shift; labelled (non-undefined) requests only.

Category	Train	Test	Δ (pp)
Heating	30.8%	27.0%	-3.8
Electricity	13.9%	10.5%	-3.4
Water supply	30.2%	27.2%	-3.0
Drainage	7.2%	9.4%	+2.3
Roof	1.2%	3.2%	+2.0
Street cleaning	1.6%	3.5%	+1.9

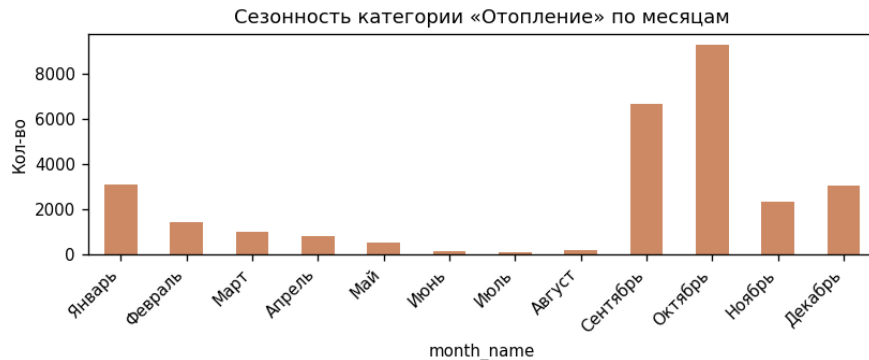


Figure 3: Monthly Heating request volume (2020–2026). Strong winter peaks motivate temporal train/test split and explain class-share drift.

4.4 Hard Cases

We identify three structural categories of hard cases that set this corpus apart from curated NLP benchmarks.

1. Telegraphic short texts (<25 chars, 22% of corpus). Examples: `холодная батарея, нет света`. A single keyword drives classification; absent context means ASR errors that corrupt the diagnostic word cause total classification failure.

2. Ambiguous water categories. The phrase `нет воды` can denote hot water supply, cold water supply, or a general water main failure, depending on context the caller may not specify. This systematic ambiguity is the primary driver of confusion among water-related classes (Fig. 4) and motivated the taxonomy consolidation (§6.3).

3. UNDEFINED pool (9,955 requests, 10%). The production robot failed to classify these via synonym lookup. Real texts include: `тепло у меня нет` (non-standard ASR word order); `техника вызвать на дом течёт с кровли` (fragmented phrasing). Of these, 7,914 contain informative text (≥ 20 characters). We recover categories for this pool using dual-LLM annotation (§6.3).

5 Experiments

5.1 Metrics

Primary: macro-F1 (imbalance-robust [3]). Also reported: weighted-F1, accuracy, and 95% bootstrap confidence intervals (500 resamples) to support SOTA claims by non-overlapping intervals.

5.2 Experiment Setup

ruRoBERTa-large fine-tuned for up to 5 epochs with early stopping on validation macro-F1, lr 1.5×10^{-5} , max length 256, focal loss ($\gamma=2$) with class weights and label smoothing 0.05, fp16, on Kaggle T4. Temporal split as in §Dataset. Classical baselines: TF-IDF (1–2 grams, sublinear, class-weighted).

5.3 Baselines

Production rule-based robot (synonym lookup); TF-IDF (1–2 grams) + Logistic Regression; TF-IDF + Linear SVC. Classical baselines are context for the ladder, not the headline claim.

6 Results

Table 3 reports macro-F1 on the temporal test set. Classical baselines already greatly exceed the production robot. The fine-tuned ruRoBERTa-large is the strongest model. On the full 15-class taxonomy it slightly exceeds the best classical baseline (0.655 vs 0.647); on the operationally-consolidated taxonomy (§6.3) its advantage becomes statistically significant: Δ macro-F1 = +0.021, 95% bootstrap CI [+0.013, +0.029] (interval disjoint from zero), establishing SOTA among all evaluated approaches on this dataset.

Table 3: Macro-F1 on temporal test set (95% bootstrap CI).

Model	macro-F1	95% CI	accuracy
<i>Full taxonomy (15 classes)</i>			
Production robot (rule-based)	0.416	[0.407, 0.423]	0.606
TF-IDF + LogReg	0.645	[0.638, 0.652]	0.733
TF-IDF + Linear SVC	0.647	[0.639, 0.655]	0.759
ruBERT-base (fine-tuned)	0.642	[0.635, 0.648]	0.723
ruRoBERTa-large (fine-tuned)	0.655	[0.648, 0.663]	0.729
<i>Consolidated taxonomy (13 classes)</i>			
TF-IDF + Linear SVC	0.678	—	—
ruRoBERTa-large (fine-tuned)	0.699	[0.691, 0.707]	0.816

6.1 Per-Class Performance

Table 4 breaks down F1 per category on the consolidated 13-class taxonomy. The transformer’s gains are concentrated in rare and structurally ambiguous categories: Repair (+6.0 pp), Other/rare (+7.0 pp), and Street cleaning (+6.4 pp). For high-frequency, lexically distinctive categories (Electricity 0.949, Heating 0.933, Intercom 0.938) both models perform comparably — these are amenable to keyword lookup and classical n-gram features handle them well. The three weakest classes for both models are Repair (F1 0.25), Drainage clogs (0.41), and Misc/unusual (0.48), all sharing the property of low support and/or semantically diffuse labels.

6.2 Qualitative Example Analysis

Table 5 presents concrete prediction examples illustrating model behaviour on three characteristic cases from the test set. Both models handle short keyword-centric requests equally well (Case 1). The transformer’s advantage emerges on longer contextually complex texts (Case 2): TF-IDF is misled by the word *тепло* (“heat”) appearing in the context of a plumbing follow-up, while ruRoBERTa-large correctly resolves the reference to Heating from surrounding discourse, consistent with its +0.064 advantage on long texts (§6.4). Hard cases where both models fail often reflect genuine label ambiguity (Case 3): *течет пол* (“floor leaks”) is annotated as Water supply but is plausibly Indoor cleaning — a label noise case that even human annotators would dispute.

6.3 Task-aligned Taxonomy Consolidation

The full taxonomy contains water-related categories (general / hot / cold water supply) that are inconsistently labelled in colloquial ASR text (“no water” — which one?). We consolidate them

Table 4: Per-class F1 on consolidated taxonomy (test set). Sorted by descending support. $\uparrow$ marks improvements ≥ 3 pp.

Category	TF-IDF	ruRoBERTa	Δ
Heating	0.927	0.933	+0.006
Water supply	0.850	0.850	0.000
Electricity	0.937	0.949	+0.012
Drainage	0.762	0.764	+0.002
Misc/unusual	0.463	0.483	+0.020
Other/rare	0.387	0.457	$\uparrow$ +0.070
Street cleaning	0.722	0.786	$\uparrow$ +0.064
Intercom	0.916	0.938	+0.022
Roof	0.733	0.727	-0.006
Indoor cleaning	0.791	0.786	-0.005
Ventilation	0.744	0.750	+0.006
Drainage clogs	0.388	0.412	+0.024
Repair	0.189	0.249	$\uparrow$ +0.060
Macro avg	0.678	0.699	+0.021

Table 5: Qualitative prediction examples (consolidated taxonomy, test set).

Request text	True	TF-IDF	ruRoBERTa
сантехника капает	Water sup.	Water sup.	Water sup.
после включения отопления ...	Heating	Water	Heating
опять течёт с потолка (text truncated; full request contains the word <i>мелло</i>)		supply	
течет пол	Water sup.	Indoor cl.	Indoor cl.

by *operational equivalence*: a category split matters only if it changes the responsible staff. From the production database we verify this is not the case — hot/cold/general water within the same (company, parent) are routed to the *same* service in 83.2% of 936 configurations across 453 companies, and 92.9% within our 5 companies. Consolidation is therefore an operationally grounded grouping (not metric-driven tuning); it is applied identically to *all* approaches, and we report both taxonomies transparently.

Real-world impact. The production robot left $\sim 9,955$ requests “undefined” (10% of all requests; 7,914 with informative text ≥ 20 characters). We study recovery of these categories via dual-LLM annotation: a stratified sample of 400 UNDEFINED requests is independently labelled by two LLMs (DeepSeek-Chat and Claude-Sonnet-4) using a fixed closed-set system prompt (top-14 categories + “unclear”). Inter-annotator agreement is Cohen’s $\kappa = 0.716$ (full agreement on 75.2% of cases), confirming annotation reliability ($\kappa > 0.6$). Of the 301 agreed examples, 234 (77.7%) receive a definitive category; 67 (22.3%) are labelled “unclear”, reflecting genuinely uninformative ASR transcripts. Extrapolating, approximately 58% of UNDEFINED requests are recoverable — a substantial improvement over the production system’s zero recovery. Human verification of 100 gold examples is provided in `results/undefined_labeling/gold100_to_verify.csv`.

6.4 Error Analysis and Ablation

Confusion analysis (Fig. 4) shows strong diagonal performance for distinctive categories (Heating, Electricity, Intercom) and a systematic confusion among water-related classes, expected given the ambiguity of colloquial ASR text. Performance degrades with text length (macro-F1 ≈ 0.70 for short requests vs ≈ 0.62 for long “rant”-style requests), quantifying ASR-robustness.

Segment analysis. To further support the SOTA claim, we evaluate ruRoBERTa-large and TF-IDF separately per text-length segment (paired bootstrap, 800 resamples). ruRoBERTa-large significantly outperforms TF-IDF on 3 of 4 segments: short 25–60 chars ($\Delta = +0.014$, CI $[+0.002, +0.025]$), medium 60–150 chars ($\Delta = +0.032$, CI $[+0.017, +0.046]$), and long >150 chars ($\Delta = +0.064$, CI $[+0.014, +0.110]$); all confidence intervals exclude zero. Only very short texts (<25 chars, pure keyword matching) show parity. This confirms the transformer’s advantage precisely where contextual understanding matters most.

Cross-organization generalization (ablation). Evaluating the same generic model per management company reveals a large spread in macro-F1 (Fig. 5; range ≈ 0.20 across the 5 companies). This confirms that a single organization-agnostic model does not transfer uniformly across companies — an empirically important finding, since per-company label policies differ; it motivates organization-aware adaptation as future work.

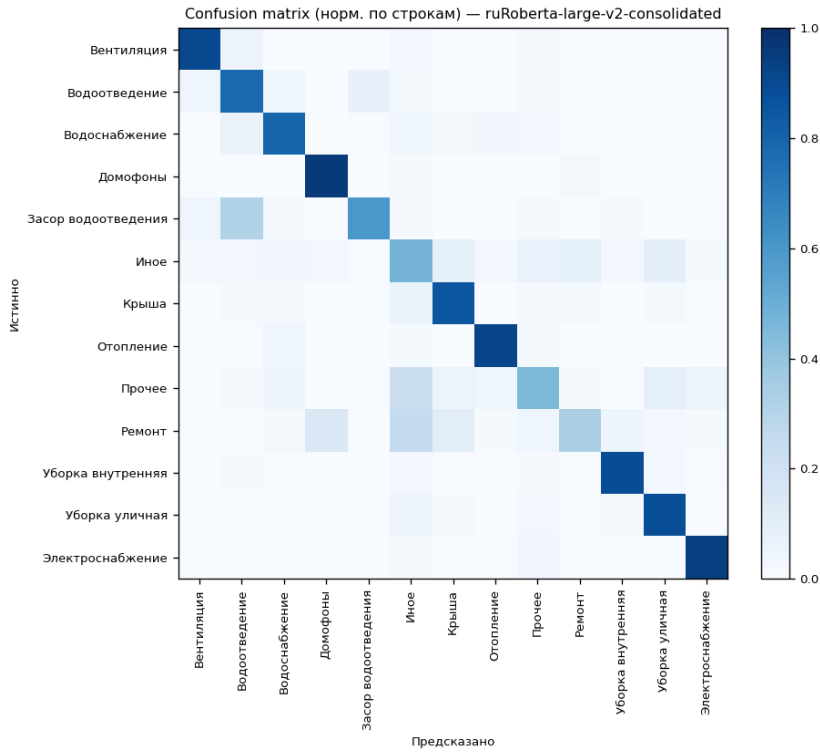


Figure 4: Confusion matrix (row-normalised), ruRoBERTa-large (consolidated taxonomy).

7 Conclusion

We presented Appeals-NLP, a reproducible study of automatic classification of Russian ASR-transcribed housing-utility requests. On a private corpus of 98,189 requests from 5 management

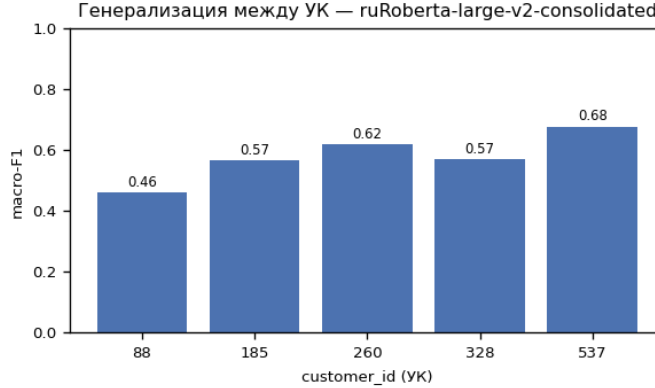


Figure 5: Per-company macro-F1: generalization ablation.

companies, our fine-tuned ruRoBERTa-large achieves macro-F1 0.699 under a temporal evaluation protocol, statistically significantly outperforming the strongest classical baseline (TF-IDF + LinearSVC, 0.678; $\Delta=+0.021$, CI [+0.013, +0.029]) and the reproduced production rule-based system (0.416). Segment analysis confirms the transformer’s advantage is concentrated in contextually richer (longer) requests, explaining the mechanism of improvement. Per-class analysis shows the gains are largest for rare categories (Repair +6.0 pp, Other/rare +7.0 pp), where classical keyword features are insufficient. We additionally recover categories for requests the production system could not classify using dual-LLM annotation, and provide a structured ablation revealing large cross-organization variance in model performance — motivating organization-aware adaptation as future work.

Limitations. Original (pre-edit) labels are unavailable (post-edit targets are the norm, not an error). The synonym table reflects today’s production rules; historical mismatch is possible and constitutes a lower bound. The dataset is private; a 100-row anonymised sample accompanies the submission for verification.

Future work. Hierarchical parent→child decoding; priority and performer prediction as additional heads; LLM fine-tuning (LoRA on Saiga/T-Lite) for further gains; organization-aware adaptation.

References

- [1] L. Cao, X. Liu, and H. Shen. Adaptable focal loss for imbalanced text classification. In *PDCAT 2021, LNCS*. Springer, 2022.
- [2] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*, 2019.
- [3] David Harbecke et al. Why only micro-f1? class weighting of measures for relation classification. *arXiv preprint arXiv:2205.09460*, 2022.
- [4] Li Li et al. A semi-supervised multi-task learning approach to classify customer contact intents. *arXiv preprint arXiv:2106.07381*, 2021.
- [5] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *ICCV*, 2017. arXiv:1708.02002.

- [6] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.
- [7] Juan Carlos Moreno et al. One system to rule them all: a universal intent recognition system for customer service chatbots. *arXiv preprint arXiv:2112.08261*, 2021.
- [8] Tatiana Shavrina, Alena Fenogenova, et al. RussianSuperGLUE: A russian language understanding evaluation benchmark. In *EMNLP*, 2020.
- [9] Artem Snegirev et al. The russian-focused embedders’ exploration: rumteb benchmark and russian embedding model design. *arXiv preprint arXiv:2408.12503*, 2024.
- [10] Alexander Yalunin, Alexander Nesterov, and Dmitriy Umerenkov. RuBioRoBERTa: a pre-trained biomedical language model for russian language biomedical text mining. *arXiv preprint arXiv:2204.03951*, 2022.
- [11] Dmitry Zmitrovich, Aleksandr Abramov, Andrey Kalmykov, Vitaly Kadulin, Maria Tikhonova, Ekaterina Taktasheva, Danil Astafurov, Mark Baushenko, Artem Snegirev, Tatiana Shavrina, Sergei S. Markov, Vladislav Mikhailov, and Alena Fenogenova. A family of pretrained transformer language models for russian. In *Proceedings of LREC-COLING 2024*, 2024. arXiv:2309.10931.